

Qualification Pack



Automotive Dealership Data Science Specialist

QP Code: ASC/Q1438

Version: 1.0

NSQF Level: 6

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ASC/Q1438: Automotive Dealership Data Science Specialist

Brief Job Description

The individual at this job is responsible for designing and developing the visualization platforms for end-to-end visibility, after-sales analytics solutions and services, and customer retention strategies based on the customer database. He/She leads the development activities and guide the team on the technical front in analytics solutions and reviewing of code and supports a range of analytical, visualization, and predictive modelling projects along with project management.

Personal Attributes

The person should have good technical and analytical skills, should have excellent interpersonal skills, communication, and presentation skills, and a good team player. The person should have project management skills, and also carry out prioritization of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9813: Manage work and resources \(Service\)](#)
2. [DGT/VSQ/N0104: Employability Skills \(120 Hours\)](#)
3. [ASC/N1465: Manage data extraction from the dealership entities](#)
4. [ASC/N1466: Prepare and analyse dealerships data by using analytical tools](#)
5. [ASC/N1467: Develop solutions for complex business problems in dealerships](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India
NSQF Level	6
Credits	22

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/2521.0100
Minimum Educational Qualification & Experience	B.E./B.Tech with 1 Year of experience in relevant field OR Certificate-NSQF (Automotive Dealership Data Analysis Engineer Level 6/ Four wheeler Service Lead Technician Level 5) with 2 Years of experience in relevant field OR M.Tech (Pursuing 2nd year) OR M.E. (Pursuing 2nd year)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	23/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-06-AU-00570-2023-V1-ASDC
NQR Version	1.1

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ASC/N9813: Manage work and resources (Service)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Ensure work as per quality standards
- Material/energy/electricity conservation practices
- Effective waste management/recycling practices
- Ensure a healthy and hygienic workplace

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** ensure that the team complies with organisation's health, safety, security policies and procedures
- PC2.** identify the risks and hazards associated with work activities, their causes and prevention as per organisation's policy
- PC3.** encourage team to report any identified breaches in health, safety, and security policies and procedures to the designated person

Ensure work as per quality standards

To be competent, the user/individual on the job must be able to:

- PC4.** ensure work area is kept clean and tidy
- PC5.** identify individual work requirements and provide necessary instructions to the team
- PC6.** ensure the team works as per the assigned and agreed requirements
- PC7.** identify work which fails the requirements, specified quality standards and ensure timely corrective action is taken
- PC8.** implement ways and guide the team to manage time, resources and cost effectively
- PC9.** train the team on skill level advancement to develop expertise in their work
- PC10.** ensure that the team understands accountability for timely completion of tasks
- PC11.** analyse and validate the problem accurately and communicate different possible solutions to the problem

Material/energy/electricity conservation practices

To be competent, the user/individual on the job must be able to:

- PC12.** identify ways to optimize usage of electricity/other source of energy and material including water in various tasks/activities/processes
- PC13.** ensure that the team uses resources in a responsible manner

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- PC14.** ensure that the team periodically checks for spills/leakages around the work area and take corrective actions or escalate to appropriate authority if unable to rectify
- PC15.** supervise team to carry out routine cleaning of tools, machine and equipment
- PC16.** ensure that the team periodically checks if the equipment/machines are maintained and functioning normally before commencing work and take corrective action wherever required

Effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

- PC17.** identify recyclable, non-recyclable and hazardous waste generated
- PC18.** ensure the team segregates waste into different categories
- PC19.** ensure proper disposal of non-recyclable waste
- PC20.** ensure recyclable and reusable material is deposited at identified location
- PC21.** ensure the team follows processes specified for disposal of hazardous waste

Ensure a healthy and hygienic workplace

To be competent, the user/individual on the job must be able to:

- PC22.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC23.** promote awareness about hygiene and sanitation regulations
- PC24.** check availability of running water, hand wash and alcohol-based sanitizers at workplace
- PC25.** support employees to cope with stress, anxiety etc.
- PC26.** wear and dispose PPEs regularly and appropriately

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of health and safety hazards that can be found in the workplace, risks and threats based on the nature of work
- KU2.** company defined workplace hazards and rules/regulation for maintaining health, safety and security at workplace
- KU3.** breaches in health, safety and security as well as procedures to report the same
- KU4.** workshop layout with electrical, hydraulic and thermal equipment used
- KU5.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU6.** ways of time and cost management
- KU7.** ways to manage efficient utilisation of energy, material and water in the process
- KU8.** ways to recognize common electrical problems and common practices of conserving electricity
- KU9.** usage of different colours of dustbins and categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU10.** organisations procedures for minimizing waste
- KU11.** waste management and methods of waste disposal
- KU12.** common sources of pollution and ways to minimize it
- KU13.** different ways for skill level advancement to develop expertise

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- KU14.** key performance indicators for the new tasks
- KU15.** timelines and goals set by the manager
- KU16.** importance of quality and timely delivery of the product/service
- KU17.** organisation's policies to maintain personal health and hygiene at workplace
- KU18.** significance of greening

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues and customers
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** complete written work with attention to detail
- GS7.** modify work practices to improve them
- GS8.** work with supervisors/team members to carry out work related tasks
- GS9.** complete tasks efficiently and accurately within stipulated time
- GS10.** make timely decisions for efficient utilization of resources
- GS11.** be punctual and utilize time
- GS12.** evaluate strategies to maintain, enhance or reduce the intensity of heightened emotional response

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	7	5	-	4
PC1. ensure that the team complies with organisation's health, safety, security policies and procedures	2	2	-	1
PC2. identify the risks and hazards associated with work activities, their causes and prevention as per organisation's policy	3	2	-	2
PC3. encourage team to report any identified breaches in health, safety, and security policies and procedures to the designated person	2	1	-	1
<i>Ensure work as per quality standards</i>	15	8	-	5
PC4. ensure work area is kept clean and tidy	2	1	-	-
PC5. identify individual work requirements and provide necessary instructions to the team	2	1	-	1
PC6. ensure the team works as per the assigned and agreed requirements	1	1	-	-
PC7. identify work which fails the requirements, specified quality standards and ensure timely corrective action is taken	3	2	-	2
PC8. implement ways and guide the team to manage time, resources and cost effectively	2	-	-	-
PC9. train the team on skill level advancement to develop expertise in their work	2	1	-	1
PC10. ensure that the team understands accountability for timely completion of tasks	2	-	-	-
PC11. analyse and validate the problem accurately and communicate different possible solutions to the problem	1	2	-	1
<i>Material/energy/electricity conservation practices</i>	10	6	-	4

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify ways to optimize usage of electricity/other source of energy and material including water in various tasks/activities/processes	2	2	-	2
PC13. ensure that the team uses resources in a responsible manner	2	1	-	-
PC14. ensure that the team periodically checks for spills/leakages around the work area and take corrective actions or escalate to appropriate authority if unable to rectify	2	1	-	1
PC15. supervise team to carry out routine cleaning of tools, machine and equipment	2	1	-	-
PC16. ensure that the team periodically checks if the equipment/machines are maintained and functioning normally before commencing work and take corrective action wherever required	2	1	-	1
<i>Effective waste management/recycling practices</i>	10	6	-	4
PC17. identify recyclable, non-recyclable and hazardous waste generated	2	2	-	1
PC18. ensure the team segregates waste into different categories	2	1	-	1
PC19. ensure proper disposal of non-recyclable waste	2	1	-	-
PC20. ensure recyclable and reusable material is deposited at identified location	2	1	-	1
PC21. ensure the team follows processes specified for disposal of hazardous waste	2	1	-	1
<i>Ensure a healthy and hygienic workplace</i>	8	5	-	3
PC22. ensure workplace, equipment, restrooms etc. are sanitized regularly	2	1	-	-
PC23. promote awareness about hygiene and sanitation regulations	2	1	-	1
PC24. check availability of running water, hand wash and alcohol-based sanitizers at workplace	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. support employees to cope with stress, anxiety etc.	1	1	-	1
PC26. wear and dispose PPEs regularly and appropriately	2	1	-	1
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9813
NOS Name	Manage work and resources (Service)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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DGT/VSQ/N0104: Employability Skills (120 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress
- PC5.** follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC6.** follow and promote environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC7.** recognize the significance of 21st Century Skills for employment

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- PC8.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC9.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC10.** use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts
- PC11.** speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front
- PC12.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC13.** write short messages, notes, letters, e-mails etc., using accurate English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC14.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC15.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC16.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC17.** use active listening techniques for effective communication
- PC18.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC19.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC20.** • ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration
- PC21.** empathize with a PwD and aid a PwD, if asked
- PC22.** escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC23.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC24.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC25.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC26.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

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To be competent, the user/individual on the job must be able to:

- PC27.** operate digital devices and use their features and applications securely and safely
- PC28.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC29.** display responsible online behaviour while using various social media platforms
- PC30.** create a personal email account, send and process received messages as per requirement
- PC31.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC32.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of Entrepreneurship and Enterprises
- PC34.** use research and networking skills to identify and assess opportunities for potential business
- PC35.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC36.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC37.** identify different types of customers
- PC38.** identify and respond to customer requests and needs in a professional manner
- PC39.** use appropriate tools to collect customer feedback
- PC40.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC41.** create a professional Curriculum vitae (Résumé)
- PC42.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC43.** apply to identified job openings using offline /online methods as per requirement
- PC44.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC45.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills
- KU2.** different learning and employability related portals
- KU3.** various constitutional and personal values
- KU4.** different environmentally sustainable practices and their importance
- KU5.** Twenty first (21st) century skills and their importance

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- KU6.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU7.** importance of career development and setting long- and short-term goals
- KU8.** Do's and don'ts of effective communication
- KU9.** POSH Act
- KU10.** inclusivity and its importance
- KU11.** different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD
- KU12.** different types of financial institutes, products, and services
- KU13.** components of salary and how to compute income and expenditure
- KU14.** importance of maintaining safety and security in offline and online financial transactions
- KU15.** different legal rights and laws
- KU16.** different types of digital devices and the procedure to operate them safely and securely
- KU17.** how to create and operate an e- mail account
- KU18.** use applications such as word processors, spreadsheets etc.
- KU19.** different types of Enterprises and ways to identify business opportunities
- KU20.** types and needs of customers
- KU21.** how to apply for a job and prepare for an interview
- KU22.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	2	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress	-	-	-	-
PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. follow and promote environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	3	-	-
PC7. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC9. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts	-	-	-	-
PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front	-	-	-	-
PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC13. write short messages, notes, letters, e-mails etc., using accurate English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC14. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC15. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	3	-	-
PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC17. use active listening techniques for effective communication	-	-	-	-
PC18. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC19. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC20. ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration	-	-	-	-
PC21. empathize with a PwD and aid a PwD, if asked	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-
PC25. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	2	3	-	-
PC27. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC29. display responsible online behaviour while using various social media platforms	-	-	-	-
PC30. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC32. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. identify different types of Entrepreneurship and Enterprises	-	-	-	-
PC34. use research and networking skills to identify and assess opportunities for potential business	-	-	-	-
PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC37. identify different types of customers	-	-	-	-
PC38. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC39. use appropriate tools to collect customer feedback	-	-	-	-
PC40. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	4	-	-
PC41. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC43. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC44. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0104
NOS Name	Employability Skills (120 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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ASC/N1465: Manage data extraction from the dealership entities

Description

This NOS unit is about performing tasks related to the collection and extraction of data from various dealership entities for answering relevant business questions.

Scope

The scope covers the following :

- Monitor assessment of project requirements
- Perform and monitor designing of project outline
- Support in selection of data integration platform to integrate the data from various department

Elements and Performance Criteria

Assessing project requirements

To be competent, the user/individual on the job must be able to:

- PC1.** Support during evaluation of the project requirements to be catered with either visualization platforms or analytics and predictive modelling solutions
- PC2.**
 - Monitor and guide team during designing of data architecture for collection and extraction of data from
 - various departments using connectors and platforms
- PC3.** Identify the people required to execute the business analytics project requirements
- PC4.** Prepare the technology stack for the front end and back end of the analytics solution
- PC5.** Assess all organizational processes related to the use of data and analytics
- PC6.** Check that the existing setup is capable or not for data collection and analysis

Perform and monitor designing of project outline

To be competent, the user/individual on the job must be able to:

- PC7.** Support and prepare an outline of the project execution taking the business questions into consideration
- PC8.** Identify appropriate data attributes to be extracted from various departments
- PC9.** Prepare the timeline and resource requirements
- PC10.** Manage project by using appropriate project tracking tools and task prioritization for all team members
- PC11.**
 - Obtain the necessary approvals within the organization for data collection and extraction from various
 - departments

Support in selection of data integration platform to integrate the data from various department

To be competent, the user/individual on the job must be able to:

- PC12.**
 - Survey and identify the existing data integration platforms considering the application integration, data
 - integration and API (Application Program Interface) management criterion.

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- PC13.** • Select the data integration platform with the capabilities like- data transformation, application connectors, file
• processing, routing, orchestration, event handling, stream processing, API management, no-vendor lock-in.
- PC14.** Design and create a data warehouse for easy consumption of data points for data analysts
- PC15.** Develop data pipelines using connectors to populate the data in the data warehouse

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining databases
- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting databases architectures and backup mechanisms
- KU4.** Who to involve while designing and developing the database architecture and pipelines for the solution
- KU5.** Range of standard platforms and tools available and how to use them
- KU6.** Database connectors and application connectors for application-cloud communications
- KU7.** Updated internal and external cybersecurity regulations
- KU8.** Data warehouse fundamentals, planning, processes, schemes, terms and SQL queries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively at the workplace
- GS2.** attentively listen and comprehend the information given by the process managers
- GS3.** write observations and any work-related information in English/regional language
- GS4.** recognise a workplace problem and take suitable action
- GS5.** • analyse and apply the information gathered from observation, experience, reasoning or communication to
• act efficiently
- GS6.** complete the assigned tasks in a timely and efficient manner
- GS7.** coordinate with shop floor workers and team for installing the new systems efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assessing project requirements</i>	15	15	-	7
PC1. Support during evaluation of the project requirements to be catered with either visualization platforms or analytics and predictive modelling solutions	3	3	-	1
PC2. • Monitor and guide team during designing of data architecture for collection and extraction of data from • various departments using connectors and platforms	2	2	-	1
PC3. Identify the people required to execute the business analytics project requirements	2	2	-	1
PC4. Prepare the technology stack for the front end and back end of the analytics solution	2	2	-	1
PC5. Assess all organizational processes related to the use of data and analytics	2	2	-	1
PC6. Check that the existing setup is capable or not for data collection and analysis	4	4	-	2
<i>Perform and monitor designing of project outline</i>	16	16	-	9
PC7. Support and prepare an outline of the project execution taking the business questions into consideration	3	3	-	2
PC8. Identify appropriate data attributes to be extracted from various departments	3	3	-	2
PC9. Prepare the timeline and resource requirements	3	3	-	2
PC10. Manage project by using appropriate project tracking tools and task prioritization for all team members	4	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Obtain the necessary approvals within the organization for data collection and extraction from various departments	3	3	-	1
<i>Support in selection of data integration platform to integrate the data from various department</i>	9	9	-	4
PC12. Survey and identify the existing data integration platforms considering the application integration, data integration and API (Application Program Interface) management criterion.	2	2	-	1
PC13. Select the data integration platform with the capabilities like- data transformation, application connectors, file processing, routing, orchestration, event handling, stream processing, API management, no-vendor lock-in.	3	3	-	1
PC14. Design and create a data warehouse for easy consumption of data points for data analysts	2	2	-	1
PC15. Develop data pipelines using connectors to populate the data in the data warehouse	2	2	-	1
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1465
NOS Name	Manage data extraction from the dealership entities
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

ASC/N1466: Prepare and analyse dealerships data by using analytical tools

Description

This NOS unit is about performing exploratory data analysis on the data extracted, deciding which data attributes are required for analytics and analysing the extracted attributes using excel/ open-source python libraires.

Scope

The scope covers the following :

- Identify business goals for which data need to be analysed,
- Support in preparation of data
- Perform statistical analysis of data

Elements and Performance Criteria

Identify business goal for which data need to be analysed

To be competent, the user/individual on the job must be able to:

- PC1.** Define business problems and business goals which can be achieved using available datasets from the manager
- PC2.** Select the relevant source of data to define dealership business goal
- PC3.** Validate the criterion in the business problem with domain person in dealerships

Support in preparation of data

To be competent, the user/individual on the job must be able to:

- PC4.** Support and create a set of metadata for the selected dataset
- PC5.** Identify the attributes or columns in the datasets which are most significant from analysis perspective
- PC6.** Perform exploratory data analysis to check for missing or duplicate data

Perform statistical analysis of data

To be competent, the user/individual on the job must be able to:

- PC7.** Perform descriptive statistical analysis on the data by following SOP
- PC8.** Perform inferential statistics analysis on the data by following SOP
- PC9.** Prepare list of highly correlated attributes
- PC10.** Find correlation amongst the selected attributes of the data and plot their heatmap

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining databases

Qualification Pack

- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting databases architectures and backup mechanisms
- KU4.** Descriptive and Inferential statistics for creating charts and predictive analytics modelling
- KU5.** Types of data wrangling and data cleaning methods to create visualization
- KU6.** Suitable documentation of the organization for the metadata creation
- KU7.** Aggregate the charts to create a dashboard to address the business problem
- KU8.** Addition of filters and chart tips to make the dashboard interactive

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively at the workplace
- GS2.** attentively listen and comprehend the information given by the process managers
- GS3.** write observations and any work-related information in English/regional language
- GS4.** recognise a workplace problem and take suitable action
- GS5.**
 - analyse and apply the information gathered from observation, experience, reasoning or communication to
 - act efficiently
- GS6.** complete the assigned tasks in a timely and efficient manner
- GS7.** coordinate with shop floor workers and team for installing the new systems efficiently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify business goal for which data need to be analysed</i>	11	11	-	6
PC1. Define business problems and business goals which can be achieved using available datasets from the manager	3	3	-	2
PC2. Select the relevant source of data to define dealership business goal	4	4	-	2
PC3. Validate the criterion in the business problem with domain person in dealerships	4	4	-	2
<i>Support in preparation of data</i>	16	16	-	7
PC4. Support and create a set of metadata for the selected dataset	6	6	-	3
PC5. Identify the attributes or columns in the datasets which are most significant from analysis perspective	5	5	-	2
PC6. Perform exploratory data analysis to check for missing or duplicate data	5	5	-	2
<i>Perform statistical analysis of data</i>	13	13	-	7
PC7. Perform descriptive statistical analysis on the data by following SOP	3	3	-	2
PC8. Perform inferential statistics analysis on the data by following SOP	3	3	-	2
PC9. Prepare list of highly correlated attributes	3	3	-	1
PC10. Find correlation amongst the selected attributes of the data and plot their heatmap	4	4	-	2
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1466
NOS Name	Prepare and analyse dealerships data by using analytical tools
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

ASC/N1467: Develop solutions for complex business problems in dealerships

Description

This NOS unit is about developing machine learning models using the extracted data and making predictive analytics solutions for complex business problems in dealerships. It also involves project management using project tracking tools and task prioritization for all team members

Scope

The scope covers the following :

- Execute training phase in the machine learning project lifecycle
- Execute testing phase in the machine learning project lifecycle
- Deployment of the developed analytics model solution into sales and support
- Deployment of the analytics model solution into production line

Elements and Performance Criteria

Execute training phase in the machine learning project lifecycle

To be competent, the user/individual on the job must be able to:

- PC1.** Select and install relevant libraries and tools for model making
- PC2.** Split and prepare the dataset into training, validation and testing sets
- PC3.**
 - Configure hyperparameters for the selected model, establish the training pipelines and execute the training
 - phase
- PC4.** Store the model and network parameters to be used in the testing phase
- PC5.** Prevent underfitting and overfitting of the model
- PC6.** Solve the imbalanced dataset problem when the samples from minority class are very few
- PC7.** Evaluate the training performance of the machine learning model for training and validation accuracy

Execute testing phase in the machine learning project lifecycle

To be competent, the user/individual on the job must be able to:

- PC8.** Test the models with testing datasets
- PC9.** Ensure the inference time per sample is as per the business requirement
- PC10.** . Evaluate the testing performance of the machine learning model for testing accuracy

Deployment of the developed analytics model solution into sales and support

To be competent, the user/individual on the job must be able to:

- PC11.** Develop a front-end application to fetch inputs from the user and consume developed model for inference
- PC12.** Verify the sales and support performance of the machine learning model
- PC13.** Analyse performance of the machine learning model and prepare feedback on the wrong predictions

Qualification Pack

PC14. Implement the feedback back to the training phase and retrain the machine learning model

Deployment of the analytics model solution into production line

To be competent, the user/individual on the job must be able to:

PC15. • Select relevant libraries and machine learning operations (MLOPS) tools and packages for deploying the
• analytics model solution into sales and support

PC16. Install the selected libraries and tools for machine learning operations tasks

PC17. • Setup the continuous training, continuous integration, and continuous delivery pipelines for the developed
• machine learning models

PC18. Monitor the analytics model solution performance in the deployment phase

PC19. Carry out commissioning of the end-to-end system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies, procedures, and guidelines that relate to designing and maintaining databases

KU2. Organizational policies and procedures for sharing data

KU3. Organizational policies and procedures for documenting databases architectures and backup mechanisms

KU4. Who to involve while developing the different stages in the machine learning lifecycle

KU5. Descriptive and inferential statistics for creating charts and predictive analytics modelling

KU6. Types of data wrangling and data cleaning methods to create visualization

KU7. Suitable documentation of the organization for the metadata creation

KU8. Updated internal and external network regulations

KU9. How to perform network assessments

KU10. How to diagnose and resolve underfitting, overfitting and imbalanced dataset issues

KU11. • How to use different machine learning algorithms for specific functions like regression, classification and
• clustering

KU12. • How to use python programming constructs for developing machine learning models using open-source
• libraries like for example, scikit-learn

KU13. How to develop necessary front end to consume the developed analytics solution

KU14. How to use MLOPS libraries and frameworks to ensure continuous training and deployment.

KU15. How to implement MLOPS in a cloud-based platform if required

KU16. How to make an API interface for the developed machine learning model in Python using REST API framework

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively at the workplace
- GS2.** attentively listen and comprehend the information given by the process managers
- GS3.** write observations and any work-related information in English/regional language
- GS4.** recognise a workplace problem and take suitable action
- GS5.**
 - analyse and apply the information gathered from observation, experience, reasoning or communication to
 - act efficiently
- GS6.** complete the assigned tasks in a timely and efficient manner
- GS7.** coordinate with shop floor workers and team for installing the new systems efficiently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Execute training phase in the machine learning project lifecycle</i>	16	16	-	7
PC1. Select and install relevant libraries and tools for model making	2	2	-	1
PC2. Split and prepare the dataset into training, validation and testing sets	3	3	-	1
PC3. • Configure hyperparameters for the selected model, establish the training pipelines and execute the training • phase	3	3	-	1
PC4. Store the model and network parameters to be used in the testing phase	2	2	-	1
PC5. Prevent underfitting and overfitting of the model	2	2	-	1
PC6. Solve the imbalanced dataset problem when the samples from minority class are very few	2	2	-	1
PC7. Evaluate the training performance of the machine learning model for training and validation accuracy	2	2	-	1
<i>Execute testing phase in the machine learning project lifecycle</i>	5	5	-	3
PC8. Test the models with testing datasets	2	2	-	1
PC9. Ensure the inference time per sample is as per the business requirement	1	1	-	1
PC10. . Evaluate the testing performance of the machine learning model for testing accuracy	2	2	-	1
<i>Deployment of the developed analytics model solution into sales and support</i>	8	8	-	4
PC11. Develop a front-end application to fetch inputs from the user and consume developed model for inference	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Verify the sales and support performance of the machine learning model	2	2	-	1
PC13. Analyse performance of the machine learning model and prepare feedback on the wrong predictions	2	2	-	1
PC14. Implement the feedback back to the training phase and retrain the machine learning model	2	2	-	1
<i>Deployment of the analytics model solution into production line</i>	11	11	-	6
PC15. • Select relevant libraries and machine learning operations (MLOPS) tools and packages for deploying the • analytics model solution into sales and support	2	2	-	1
PC16. Install the selected libraries and tools for machine learning operations tasks	2	2	-	1
PC17. • Setup the continuous training, continuous integration, and continuous delivery pipelines for the developed • machine learning models	3	3	-	2
PC18. Monitor the analytics model solution performance in the deployment phase	2	2	-	1
PC19. Carry out commissioning of the end-to-end system	2	2	-	1
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1467
NOS Name	Develop solutions for complex business problems in dealerships
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at

Qualification Pack

each examination/ training center based on these criteria.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9813.Manage work and resources (Service)	50	30	-	20	100	15
DGT/VSQ/N0104.Employability Skills (120 Hours)	20	30	-	-	50	10
ASC/N1465.Manage data extraction from the dealership entities	40	40	0	20	100	25
ASC/N1466.Prepare and analyse dealerships data by using analytical tools	40	40	0	20	100	25
ASC/N1467.Develop solutions for complex business problems in dealerships	40	40	0	20	100	25
Total	190	180	-	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PC	Performance Criteria
KU	Knowledge and Understanding
GS	Generic Skills
API	Application Program Interface
SQL	Structured Query Language
NOS	National Occupational Standard(s)

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.

Qualification Pack

National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.